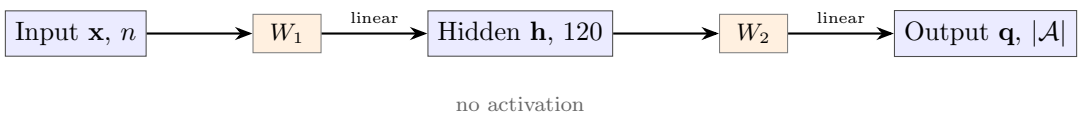
DQN 3-Layers Vectorial

1 Network

Fully linear Q-network (no activation). Architecture:



Equations:

$$\mathbf{h} = W_1\mathbf{x} + \mathbf{b}_1, \qquad \mathbf{q} = W_2\mathbf{h} + \mathbf{b}_2 \tag{1}$$

with $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{h} \in \mathbb{R}^{120}$, $\mathbf{q} \in \mathbb{R}^{|\mathcal{A}|}$. So

$$Q(s,a) = \mathbf{q}_a = (W_2W_1)_{a,:}\mathbf{x} + \text{bias}_a. \tag{2}$$

2 Feature contribution

Effective weight from input to Q per action:

$$W_{\text{eff}} = W_2W_1 \quad \in \mathbb{R}^{|\mathcal{A}|\times n}. \tag{3}$$

Contribution of feature i to action a (additive term in $Q(s,a)$):

$$c_{a,i}(s) = (W_{\text{eff}})_{a,i}\, x_i. \tag{4}$$

For chosen action a_b in a batch: $\mathbf{c}_b = (W_{\text{eff}}[a_b,:]) \odot \mathbf{x}_b$.

3 Vectorial reward

Per-feature reward (reward backpropagated to features):

$$r_{\text{vec},i} = \frac{r}{n} \quad \forall i. \tag{5}$$

So $\mathbf{r}_{\text{vec}} \in \mathbb{R}^n$ has each entry r/n.

4 Loss

TD target (no gradient):

$$y = r + \gamma \max_{a'} Q_{\text{target}}(s', a') \cdot (1 - \text{done}). \tag{6}$$

TD loss:

$$L_{\text{td}} = \frac{1}{B} \sum_{b=1}^{B} \big(y_b - Q(s_b, a_b)\big)^2. \tag{7}$$

Per-feature loss (contribution of feature i to chosen action vs r/n):

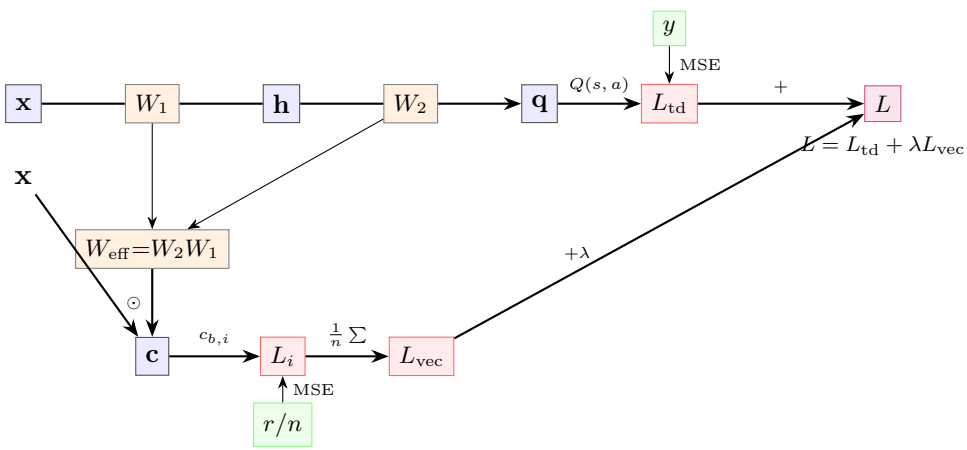
$$L_i = \frac{1}{B} \sum_{b=1}^{B} \big(c_{b,i} - \tfrac{r_b}{n}\big)^2. \tag{8}$$

Vectorial loss and total loss:

$$L_{\text{vec}} = \frac{1}{n} \sum_{i=1}^{n} L_i, \qquad L = L_{\text{td}} + \lambda\, L_{\text{vec}}. \tag{9}$$

λ is `vectorial_loss_coef` (default 0.1).

4.1 How the loss connects to the architecture



Top row: Forward pass $\mathbf{x} \to W_1 \to \mathbf{h} \to W_2 \to \mathbf{q}$; $Q(s,a)$ is the chosen action's value; L_{td} is MSE between $Q(s,a)$ and TD target y. **Bottom row:** $W_{\text{eff}} = W_2 W_1$ gives per-feature contribution $\mathbf{c} = W_{\text{eff}}[a,:] \odot \mathbf{x}$; each L_i is MSE between $c_{b,i}$ and r/n; $L_{\text{vec}} = \frac{1}{n}\sum_i L_i$. **Total:** $L = L_{\text{td}} + \lambda L_{\text{vec}}$ is backpropagated through the network to update W_1, W_2.